

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 17: Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Cyclic quadrilaterals

1. A, E, C and B lie in order on a circle. Chords AC and BC are equal, BC is produced to D , and $\angle ACD = 120^\circ$. Find $\angle AEC$. (a) 30° (b) 60° (c) 90° (d) 120° **ICSE 2025, 1 mark**

SOLUTION

$$\angle ACB = 180^\circ - 120^\circ = 60^\circ \text{ (linear pair).}$$

$$AC = BC, \text{ so } \angle ABC = \angle BAC = \frac{180^\circ - 60^\circ}{2} = 60^\circ.$$

$$ABCE \text{ is a cyclic quadrilateral, so } \angle AEC = 180^\circ - \angle ABC = 120^\circ.$$

Answer: (d)

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